6. Results and Discussion

All classical and neural models were evaluated using the same leakage-safe temporal protocol. January-July 2025 formed the training period, August was used for model selection and residual calibration, and 2,865 forecast origins in September were retained for final testing. Each origin contributed 16 quarter-hour targets, yielding 45,840 held-out values over a four-hour horizon. Within each model family, the unrestricted 72-hour and manually defined 42-hour representations used the same validation and test origins. Persistence, defined by holding the latest observed target value across the horizon, was used as the operational baseline.

Classical unrestricted models received 1,440 flattened inputs (288-time steps across five stations), whereas their manual counterpart used four physically motivated scalar features. Sequence models received either 288 x 5 unrestricted histories or 168 x 4 manual histories. The comparison therefore tests whether a compact lag-informed representation improves generalization; it does not isolate the causal effect of a single observation at exactly 42 hours.

6.1 Classical Machine Learning Performance

6.2 Sequence Networks, Foundation Model, and Ensemble

Table X shows that every neural configuration achieved positive skill.

All models were evaluated on the same 2,865 held-out September forecast origins and the same 45,840 quarter-hour target values. Consequently, the differences below reflect the learned models and their feature representations rather than different data windows.

6.1 Overall Matched-Window Performance

Table 1 summarizes performance across the complete four-hour trajectories. The explicit-42-hour XGBoost variant achieved the lowest RMSE and highest R2, while the LSTM achieved the lowest MAE. The TCN was similarly competitive. These three models improved RMSE over persistence by approximately 8.8-9.9%, whereas both SVR configurations failed to exceed the baseline.

Table 1. Overall performance across all four-hour test trajectories.

Model	MAE (mS/cm)	RMSE (mS/cm)	R2	RMSE skill (%)
XGBoost	0.1338	0.2097	0.4644	6.53
XGBoost (explicit 42 h)	0.1206	0.2022	0.5021	9.88
SVR	0.1384	0.2294	0.3592	-2.23
SVR (explicit 42 h)	0.1674	0.2487	0.2470	-10.83
LSTM	0.1172	0.2039	0.4936	9.12
TCN	0.1187	0.2046	0.4903	8.82
Persistence	0.1262	0.2244	0.3869	0.00

The identity of the nominally best model depended on the error criterion. LSTM reduced the typical absolute deviation most effectively, whereas explicit-42-hour XGBoost obtained the lowest squared-error measure, indicating stronger control of some larger errors. The pairwise RMSE differences among explicit-42-hour XGBoost, LSTM and TCN ranged from 0.0007 to 0.0024 mS/cm. They should therefore be treated as practically similar in this experiment unless uncertainty estimates over independent hydrological periods establish a stable ranking.

6.2 Lead-Time Behavior

Persistence remained a strong baseline at short lead times because conductivity is locally autocorrelated, but its relative adequacy decreased as the horizon increased. Table 2 reports RMSE skill at the one-, two-, three- and four-hour endpoints. The LSTM and TCN gains increased from approximately 5.5-5.8% at one hour mark to more than 10% at four hour mark. Explicit-42-hour XGBoost remained positive throughout and reached 11.84% at four hours. This pattern indicates that learned spatial-temporal information becomes more useful as the unchanged-value assumption progressively deteriorates.

Table 2. RMSE skill relative to persistence at selected lead times (higher is better).

Model	1 h	2 h	3 h	4 h
XGBoost	4.73	5.47	8.07	6.45
XGBoost (explicit 42 h)	9.46	8.68	9.88	11.84
SVR	-2.46	-3.07	-1.64	-1.49
SVR (explicit 42 h)	-8.48	-10.83	-11.63	-10.46
LSTM	5.50	8.38	10.05	10.59
TCN	5.80	8.42	9.42	10.29

6.3 Model-Specific Findings

~~6.3.1 XGBoost~~

~~The complete-history XGBoost model improved overall RMSE by 6.53% relative to persistence but had a higher MAE than persistence. This combination suggests that the trees reduced some large errors while introducing small or moderate corrections that were less consistently beneficial. Replacing the full history with the compact explicit-42-hour representation reduced RMSE from 0.2097 to 0.2022 mS/cm, reduced MAE from 0.1338 to 0.1206 mS/cm, and improved every reported endpoint. The advantage was largest at four hours, where RMSE decreased from 0.2637 to 0.2485 mS/cm.~~

~~The explicit representation did not dominate every validation criterion: August MAE was lower than that of full-history XGBoost, but its validation RMSE was slightly higher (0.2197 versus 0.2151 mS/cm). The September gain is therefore promising but not uniformly reproduced across months. The current matched experiment did not report a feature-importance analysis for the four compact inputs. The observed gain should therefore be attributed to the compact representation as a whole, rather than specifically to the Terneuzen value at t - 42 hours.~~

~~6.3.2 Support Vector Regression~~

~~Full-history SVR produced an RMSE of 0.2294 mS/cm, corresponding to negative skill of -2.23%. The explicit-42-hour representation deteriorated further to 0.2487 mS/cm and negative skill of -10.83%, with negative skill at every reported horizon. Thus, the compact feature scheme was beneficial to XGBoost but not to SVR. This algorithm-dependent response argues against interpreting the ablation as universal evidence for a deterministic 42-hour relationship. The SVR hyperparameters were intentionally held fixed to isolate the representation change; because gamma = 'scale' changes the effective RBF kernel width when dimensionality changes, a separately tuned compact SVR could behave differently.~~

~~6.3.3 Long Short-Term Memory Network~~

~~The LSTM achieved the lowest MAE (0.1172 mS/cm), an RMSE of 0.2039 mS/cm and R2 of 0.4936. Its RMSE skill increased monotonically from 5.50% at one hour to 10.59% at four hours. Training stopped after 60 epochs, and the best August Huber loss occurred at epoch 42. The LSTM therefore learned corrections that became increasingly valuable at longer lead times while retaining slightly lower typical absolute error than the other competitive models.~~

~~6.3.4 Temporal Convolutional Network~~

~~The TCN achieved an MAE of 0.1187 mS/cm, an RMSE of 0.2046 mS/cm and R2 of 0.4903. Its overall RMSE differed from the LSTM by only 0.0007 mS/cm. Training ran for 82 epochs, with the lowest validation Huber loss at epoch 68. Skill increased from 5.80% at one hour to 10.29% at four hours, demonstrating that causal dilated convolution extracted useful information from the unshifted 72-hour histories. Because neither neural network was given a fixed 42-hour column, their comparable accuracy shows that competitive performance did not require manually imposing the propagation hypothesis.~~

~~6.4 Interpretation of the 42-Hour Hypothesis~~

~~The explicit-42-hour XGBoost result is consistent with the proposed Terneuzen-to-Ghent timescale being potentially informative, especially at the four-hour endpoint. It is not, however, a controlled confirmation of a unique plume travel time. The full-history models already contained the same observation, and the explicit experiment simultaneously removed 1,436 other features. The improvement may therefore arise from regularization, concentration on recent Indusii momentum, or the selected Terneuzen lag. A decisive attribution would require otherwise identical compact models without Terneuzen, with alternative lags such as 24, 36 and 48 hours, and with a temporally shuffled Terneuzen control. Lag choice should be fixed using training and validation data before final testing.~~

6.5 Limitations and Operational Implications

The positive skill of explicit-42-hour XGBoost, LSTM and TCN relative to persistence is operationally relevant: each provides a measurable improvement over simply holding the latest conductivity value, and the advantage grows toward the four-hour horizon where operators have more time to respond. Nevertheless, R2 remained close to 0.5 overall, indicating that substantial variability was not explained. Example trajectories also showed smoothing toward the recent level and incomplete reproduction of rapid changes.

The evaluation covers one held-out month, while adjacent 15-minute forecast origins share most of their histories and future targets. The 2,865 test rows must not therefore be treated as independent samples for significance testing. In addition, the models excluded potentially important drivers such as tide, canal discharge, freshwater inflow, rainfall, temperature, vessel movements and lock-operation records. Future evaluation should use blocked event- or day-level uncertainty estimates, additional seasons and years, and external hydrodynamic covariates. Under the present evidence, explicit-42-hour XGBoost, LSTM and TCN are best regarded as a competitive group rather than as a definitive ranking, with the 42-hour delay retained as a physically motivated hypothesis requiring further controlled validation.